GCG Data Analyst Test

Introduction:

Take as long as you like to answer the following questions. Your answers should stand alone and not require presenting to give complete information.

The test:

The objective of this small project is to test your technical skills and understanding of game data.

The data is for a two month period from a fictional game.

In addition to your response please include any code / additional plots / workings you created or think are relevant.

The data:

You have been given a file containing csv files of event types from a fictional game. We suggest you set up a SQLlite3 database (or similar) to help you answer the questions, however, you are free to use whichever tools you would prefer.

The data contains 5 event types:

- register: sent on first launch
- session_start: sent at the end of each player session (event_time is the start of the session ie login, user attributes are the values at the end of the player session)
- revenue: sent on a real money purchase
- ad_revenue: sent after the player views a rewarded advert
- battle_short: sent at the end of a player battle (battles are available AFTER the player completes the tutorial

There is an explanation of each field at the end of this doc.

In order to complete this test, please provide your answers in a Tableau workbook.
You can download Tableau for free, save a file of your work and email it to us with any other documentation, queries etc (or push it to your personal Github repository or similar location).

Questions

Question 1:

- Which metrics do you think are most important for calculating the state and health of a mobile free-to-play game?
    - Why are they important?

Question 2

The following questions require you to create a single Tableau report for all of the questions. Add as many filters, views etc as you want to complete the tasks.

- From the data provided create a Tableau report highlighting day 1, day 3, and day 7 retention over time.
    - Note: We consider day 1 retention to be the proportion of users who return the first day **after** they register (day 0).
      We use a rolling 24 hour window for retention (this is the user_point_in_lifetime field) calculations.
- Analyse the geographic and/or platform split or any other segmentation you believe to be valuable of the revenue and the users
- Display two of the following KPI's over time (the two are entirely your own choice)
    - Installs
    - Revenue
    - LTV (RPI)
    - ARPPU
- Bonus points:
    - Help the game designer with one recommendation or insight for the campaign battles.
      NOTE: You do not need to use Tableau for this question - feel free to use any tool you prefer.
        - Event: sod_battle_short
        - Column custom_battle_type = 'campaign'

Question 3 - Visualisations:

- Calculate and visualise one of the metrics you gave in answer to Question 1a - or any other metric you feel is valuable
    - Feel free to add any other visualisations or recommendations you feel are relevant.

GARDEN CITY GAMES

Additional Information:

Name	Definition
user_id	A unique user ID
event_time	Timestamp of the event
user_point_in_lifetime	integer derived from the players days (calculated by a 24 hour window from register) in the game. First day in the game is Day 0, the second 24 hour window is day 1 etc.
user_country	The country detected via player IP when the player installed the game
user_device_platform	The store where the player downloaded the game from (iOS or Android)
user_marketing_user	Boolean. If TRUE then the user installed the game from a paid advertisement, if FALSE the player “organically” installed the game from the platform store
user_session_id	A unique session ID
event_type	Type of event
game_name	Name of the game
user_install_time	Timestamp of when the player first installed the game
user_cumulative_iap_revenue	Total player IAP spend in game
user_cumulative_iap_payments	Total number of IAP’s a player has made
user_cumulative_sessions	Total number of sessions in the players lifetime
event_session_length	Amount of time the player has spent in the current session

user_balance_gold	Users current balance of gold (hard currency) at the time the event was recorded
user_payer_user	Boolean value. TRUE if player has made at least one IAP

event_revenue_gross_usd	The revenue recorded from the event, in gross USD (before platform fees, taxes etc).
event_item_id	The item purchased (in revenue event)
event_battle_result	The result of the battle
custom_battle_type	The type of battle the player played
event_battle_duration	The length of the battle in seconds
user_cumulative_battles	Total number of battles a player has played in their lifetime
custom_map_id	The identifier of the battle map

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